

① $\lim_{n \rightarrow \infty} (1+x_n)^{y_n}$

Trudniejsze

② $\lim \geq R.H \leftarrow$ de Hospitala *

③ znaleźć obszar zb. szeregu „ $\sum x$ ”

④ pochodna skomplikowanej funkcji złożonej

⑤ Badanie przebiegu zmienności funkcji ln *



25.23 d f

25.24 e f

25.25 d k k

Zad. 25.23. d) $b > 0$

$$d) \lim_{w \rightarrow 0^+} \frac{e^{a\sqrt{w}} - 1}{\sqrt{\sin bw}} = \left[\frac{e^0 - 1}{\sqrt{\sin 0}} \right] = \left[\frac{0}{0} \right] \xrightarrow{H} \frac{e^{a\sqrt{w}} \cdot a \cdot \frac{1}{2\sqrt{w}}}{\frac{-1}{\sqrt{\sin bw}} \cdot \cos bw \cdot b} \rightarrow$$

$\rightarrow$

$$\begin{aligned} f) \lim_{x \rightarrow -\infty} \frac{e^{\frac{1}{x}} - 1}{\arctan x - \frac{\pi}{2}} &= \left[\frac{0}{0} \right] \xrightarrow{H} \frac{e^{\frac{1}{x}} \cdot \frac{-1}{x^2}}{\frac{-1}{1+x^2}} \rightarrow \frac{e^{\frac{1}{x}} (1+x^2)}{x^2 \nearrow \infty} \rightarrow \\ &\rightarrow \left[\frac{\infty}{\infty} \right] \xrightarrow{H} \frac{e^{\frac{1}{x}} \left(\frac{-1}{x^2} \right) (1+x^2) + 2x e^{\frac{1}{x}}}{2x} \rightarrow \\ &\rightarrow \frac{e^{\frac{1}{x}} \left(-\frac{1}{x^2} - 1 \right) + 2x e^{\frac{1}{x}}}{2x} \rightarrow \frac{e^{\frac{1}{x}} \left(2x - \frac{1}{x^2} - 1 \right)}{2x} \rightarrow \\ &\rightarrow e^{\frac{1}{x}} \left(1 - \frac{1}{2x^3} - \frac{1}{2x} \right) \rightarrow [1 \cdot (1 - 0 - 0)] \rightarrow 1 \end{aligned}$$

Zad. 25.24. e f

$$e) \lim_{x \rightarrow +\infty} \frac{x^2}{\ln x} = \left[\frac{\infty}{\infty} \right] \xrightarrow{H} \frac{2x}{\frac{1}{x}} \rightarrow 2x^2 \rightarrow +\infty$$

$$f) \lim_{x \rightarrow -\infty} \frac{x^2}{e^{x^2}} = \left[\frac{\infty}{e^{\infty} = \infty} \right] \xrightarrow{H} \frac{2x}{e^{x^2} \cdot 2x} \rightarrow \frac{1}{e^{x^2}} \xrightarrow{\infty} 0$$

Zad. 25.25. d h k

$$d) \lim_{t \rightarrow 0^+} (t \cdot e^{\frac{1}{t}}) = [0 \cdot e^{\frac{1}{0^+}}] = [0 \cdot +\infty] \rightarrow \frac{e^{\frac{1}{t}}}{\frac{1}{t}} \xrightarrow{H} \frac{e^{\frac{1}{t}} \cdot \frac{-1}{t^2}}{-\frac{1}{t^2}} \rightarrow e^{\frac{1}{t}} \xrightarrow{+\infty} +\infty$$

$$h) \lim_{v \rightarrow +\infty} [v(e^{\frac{1}{v}} - 1)] = [+\infty \cdot (1-1)] = [+\infty \cdot 0] \rightarrow \frac{e^{\frac{1}{v}} - 1}{\frac{1}{v}} \xrightarrow{H} \frac{e^{\frac{1}{v}} \cdot (-\frac{1}{v^2})}{-\frac{1}{v^2}} \rightarrow e^{\frac{1}{v}} \xrightarrow{0} 1$$

$$k) \lim_{x \rightarrow -\infty} [(x-1) \cdot e^{\frac{1}{x-1}} - x] = [-\infty + \infty] \rightarrow x \left(\frac{x-1}{x} \cdot e^{\frac{1}{x-1}} - 1 \right) \rightarrow \frac{\frac{x-1}{x} \cdot e^{\frac{1}{x-1}} - 1}{\frac{1}{x}} \rightarrow \frac{\left(1 - \frac{1}{x}\right) \cdot e^{\frac{1}{x-1}} - 1}{\frac{1}{x}} \rightarrow \left[\frac{(1-0) \cdot 1 - 1}{0} = \frac{0}{0} \right] \xrightarrow{H} \frac{\frac{1}{x^2} \cdot e^{\frac{1}{x-1}} + \left(1 - \frac{1}{x}\right) \cdot e^{\frac{1}{x-1}} \cdot \left(\frac{-1}{(x-1)^2}\right) \cdot 1}{-\frac{1}{x^2}} \rightarrow$$

$$\begin{aligned}
 & \xrightarrow{H} \frac{\frac{1}{x^2} \cdot e^{\frac{1}{x-1}} + \left(1 - \frac{1}{x}\right) \cdot e^{\frac{1}{x-1}} \cdot \left(\frac{-1}{(x-1)^2}\right) \cdot 1}{-\frac{1}{x^2}} \rightarrow \\
 & \rightarrow \frac{e^{\frac{1}{x-1}} \left(\frac{1}{x^2} - \left(\frac{x-1}{x}\right) \left(\frac{1}{(x-1)^2}\right) \right)}{-\frac{1}{x^2}} \rightarrow \\
 & \rightarrow \frac{e^{\frac{1}{x-1}} \left(\frac{1}{x^2} - \frac{1}{x(x-1)} \right)}{-\frac{1}{x^2}} \rightarrow \frac{e^{\frac{1}{x-1}} \left(\frac{x-1-x}{x^2(x-1)} \right)}{-\frac{1}{x^2}} \rightarrow \frac{e^{\frac{1}{x-1}} \left(\frac{-1}{x^2(x-1)} \right)}{-\frac{1}{x^2}} \rightarrow \\
 & \rightarrow \frac{e^{\frac{1}{x-1}}}{x-1} \rightarrow \left[\frac{1}{-\infty} \right] \rightarrow 0
 \end{aligned}$$

Zad. 25.26. b, c, g, l

$$b) \lim_{x \rightarrow 0^+} x^{\frac{1}{\ln(e^x-1)}} \rightarrow e^{\frac{1}{\ln(e^x-1)} \cdot \ln x} \rightarrow e^{\frac{\ln x}{\ln(e^x-1)}} \rightarrow e^1 \rightarrow e$$

$$\begin{aligned}
 \lim_{x \rightarrow 0^+} \frac{\ln x}{\ln(e^x-1)} & \rightarrow \left[\frac{-\infty}{-\infty} \right] \xrightarrow{H} \frac{\frac{1}{x}}{e^x-1 \cdot e^x} \rightarrow \frac{e^x-1}{e^x \cdot x} \rightarrow \left[\frac{0}{0} \right] \xrightarrow{H} \\
 & \xrightarrow{H} \frac{e^x}{e^x \cdot x + 1 \cdot e^x} \rightarrow \frac{1}{x+1} \rightarrow 1
 \end{aligned}$$

$$c) \lim_{x \rightarrow 0} [\ln(x+1)]^x \rightarrow e^{x \cdot \ln(\ln(x+1))} \rightarrow e^0 \rightarrow 1$$

$$\begin{aligned}
 \lim_{x \rightarrow 0} [x \cdot \ln(\ln(x+1))] & = [0 \cdot -\infty] \rightarrow \frac{\ln(\ln(x+1))}{\frac{1}{x}} \rightarrow \left[\frac{-\infty}{\infty} \right] \xrightarrow{H} \\
 & \xrightarrow{H} \frac{\frac{1}{\ln(x+1)} \cdot \frac{1}{x+1}}{-\frac{1}{x^2}} \rightarrow \frac{-x^2}{\ln(x+1) \cdot (x+1)} \rightarrow \left[\frac{0}{0} \right] \xrightarrow{H} \frac{-2x}{\frac{1}{x+1} (x+1) + \ln(x+1)} \rightarrow \\
 & \rightarrow \frac{-2x}{1 + \ln(x+1)} \rightarrow \left[\frac{0}{1+0} \right] \rightarrow 0
 \end{aligned}$$

$$g) \lim_{x \rightarrow +\infty} \left(\frac{1}{x}\right) e^{-x} \rightarrow e^{-x \ln \frac{1}{x}}$$

$$\lim_{x \rightarrow +\infty} e^{-x} \cdot \ln \frac{1}{x} \rightarrow \frac{\ln \frac{1}{x}}{e^x} \rightarrow \left[\frac{-\infty}{\infty} \right] \xrightarrow{H} \frac{\frac{1}{x} \cdot \left(-\frac{1}{x^2}\right)}{e^x} \rightarrow \frac{-\frac{1}{x}}{e^x} \rightarrow$$

$$\rightarrow -\frac{1}{x e^x} \rightarrow \left[\frac{-1}{-\infty \cdot \infty} \right] \rightarrow 0$$

$$I) \lim_{x \rightarrow +\infty} \left(\frac{2}{\pi} \arctg x\right)^{x^2} \rightarrow e^{x^2 \cdot \ln\left(\frac{2}{\pi} \cdot \arctg x\right)}$$

$$\lim_{x \rightarrow +\infty} x^2 \cdot \ln\left(\frac{2}{\pi} \cdot \arctg x\right) \rightarrow [\infty \cdot 0] \rightarrow \frac{\ln\left(\frac{2}{\pi} \arctg x\right)}{\frac{1}{x^2}} \rightarrow \left[\frac{0}{0} \right] \xrightarrow{H}$$

$$\xrightarrow{H} \frac{\frac{1}{\frac{2}{\pi} \arctg x} \cdot \frac{1}{1+x^2} \cdot \frac{2}{\pi}}{\frac{-2}{x^3}} \rightarrow \frac{-x^3}{2 \arctg x (1+x^2)} \rightarrow \left[\frac{-\infty}{2 \cdot \frac{\pi}{2} \cdot \infty} = \frac{-\infty}{\infty} \right] \xrightarrow{H}$$

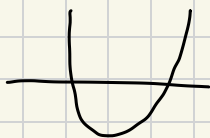
$$\rightarrow \frac{-3x^2}{2(1+2x \arctg x)} \rightarrow \left[\frac{-\infty}{\infty} \right] \xrightarrow{H} -$$

→

Zad. 25.29. d f g h

d)

$$g(x) = \frac{x^3 - x + 2}{x^2 - 6}$$



$$D: x \in \mathbb{R} \setminus \{-\sqrt{6}, \sqrt{6}\}$$

$$x = -\sqrt{6}?$$

$$\lim_{x \rightarrow -\sqrt{6}^-} \frac{x^3 - x + 2}{x^2 - 6} \rightarrow \left[\frac{\overset{LO}{(-5\sqrt{6}+2)}}{0^+} \right] \rightarrow -\infty$$

$\Rightarrow x = -\sqrt{6}$ as. pionowa
dystonna

$$\lim_{x \rightarrow -\sqrt{6}^+} \frac{x^3 - x + 2}{x^2 - 6} \rightarrow \left[\frac{\overset{LO}{-5\sqrt{6}+2}}{0^-} \right] \rightarrow +\infty$$

$$\lim_{x \rightarrow \sqrt{6}^-} \frac{x^3 - x + 2}{x^2 - 6} \rightarrow \left[\frac{5\sqrt{6}+2}{0^-} \right] \rightarrow -\infty$$

$\Rightarrow x = \sqrt{6}$ as. pionowa
dystonna

$$\lim_{x \rightarrow \sqrt{6}^+} \frac{x^3 - x + 2}{x^2 - 6} \rightarrow \left[\frac{5\sqrt{6}+2}{0^+} \right] \rightarrow +\infty$$

Asymptoty ukośne

$$a_L = \lim_{x \rightarrow -\infty} \frac{f(x)}{x} \rightarrow \frac{x^2 - 1 + \overset{LO}{\frac{2}{x}}}{x^2 - 6} \rightarrow 1 = a_p$$

$$b_L = \lim_{x \rightarrow -\infty} (f(x) - a_L x) \rightarrow \frac{x^3 - x + 2}{x^2 - 6} - x \rightarrow \frac{x^3 - x + 2 - x^3 + 6x}{x^2 - 6} \rightarrow$$

$$\rightarrow \frac{5x + 2}{x^2 - 6} \rightarrow 0 = b_p$$

$$y = x$$

$$f) w(z) = z \cdot e^{\frac{1}{z}}$$

$$D: z \in \mathbb{R} \setminus \{0\}$$

Asymptoty pionowe

$$z = 0?$$

$$\lim_{z \rightarrow 0^-} z \cdot e^{\left(\frac{1}{z}\right)^{\rightarrow -\infty}} \rightarrow 0$$

$$\lim_{z \rightarrow 0^+} z \cdot e^{\left(\frac{1}{z}\right)^{\rightarrow +\infty}} \rightarrow [0 \cdot \infty] \rightarrow \frac{e^{\frac{1}{z}}}{\frac{1}{z}} \rightarrow \left[\frac{+\infty}{+\infty} \right] \rightarrow \frac{e^{\frac{1}{z}} \cdot \left(-\frac{1}{z^2}\right)}{\left(-\frac{1}{z^2}\right)} \rightarrow e^{\frac{1}{z}} \rightarrow +\infty$$

$z = 0$ as. pionowa prawostronna

Asymptoty ukośne

$$a_L = \lim_{z \rightarrow -\infty} \frac{w(z)}{z} \rightarrow e^{\left(\frac{1}{z}\right)^{\rightarrow 0}} \rightarrow e^0 \rightarrow 1 = a_P$$

$$b_L = \lim_{z \rightarrow -\infty} (w(z) - a_L z) = \lim_{z \rightarrow -\infty} (z e^{\frac{1}{z}} - z) \rightarrow z(e^{\frac{1}{z}} - 1) \rightarrow$$

$$\rightarrow \frac{e^{\frac{1}{z}} - 1}{\frac{1}{z}} \rightarrow \left[\frac{0}{0} \right] \xrightarrow{H} \frac{e^{\frac{1}{z}} \cdot \left(-\frac{1}{z^2}\right)}{\left(-\frac{1}{z^2}\right)} \rightarrow e^{\frac{1}{z}} \rightarrow 1 = b_P$$

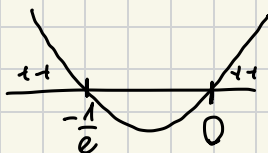
as. ukośna $y = z + 1$

g) $l(x) = x \ln(e + \frac{1}{x})$

$x \neq 0 \quad e + \frac{1}{x} > 0 \parallel \cdot x^2$

$ex^2 + x > 0$

$x(ex + 1) > 0$



D: $x \in (-\infty; -\frac{1}{e}) \cup (0; +\infty)$

Asymptoty ukośne

$a_L = \lim_{x \rightarrow -\infty} \ln(e + \frac{1}{x}) \rightarrow 1 = a_p$

$b_L = \lim_{x \rightarrow \infty} (x \ln(e + \frac{1}{x}) - x) \rightarrow$

$\rightarrow x(\ln(e + \frac{1}{x}) - 1) \rightarrow$

$\rightarrow \frac{\ln(e + \frac{1}{x}) - 1}{\frac{1}{x}} \rightarrow$

$\rightarrow \left[\frac{1-1}{0} = \frac{0}{0} \right] \xrightarrow{H}$

$\xrightarrow{H} \frac{\frac{1}{e + \frac{1}{x}} \cdot (-\frac{1}{x^2})}{(-\frac{1}{x^2})} \rightarrow \frac{1}{e + \frac{1}{x}} \rightarrow \frac{1}{e} = b_p$

as. ukośna $y = x + \frac{1}{e}$

Asymptoty pionowe

$x = 0?$

$\lim_{x \rightarrow 0^-} x \ln(e + \frac{1}{x}) = \frac{\ln(e + \frac{1}{x})}{\frac{1}{x}} \rightarrow \left[\frac{-\infty}{-\infty} \right]$

$\xrightarrow{H} \frac{\frac{1}{e + \frac{1}{x}} \cdot (-\frac{1}{x^2})}{(-\frac{1}{x^2})} \rightarrow \frac{1}{e + \frac{1}{x}} \rightarrow 0$

$\lim_{x \rightarrow 0^+} x \ln(e + \frac{1}{x}) = \frac{\ln(e + \frac{1}{x})}{\frac{1}{x}} \rightarrow \left[\frac{\infty}{\infty} \right] \rightarrow$

$\xrightarrow{H} \frac{\frac{1}{e + \frac{1}{x}} \cdot (-\frac{1}{x^2})}{(-\frac{1}{x^2})} \rightarrow \frac{1}{e + \frac{1}{x}} \rightarrow 0$

$x = 0$ nie jest asymptotą pionową

$x = -\frac{1}{e}$

$\lim_{x \rightarrow -\frac{1}{e}^-} x \ln(e + \frac{1}{x}) \rightarrow \left[-\frac{1}{e} \cdot \ln(e - e^0) \right]$

$\rightarrow \left[-\frac{1}{e} \cdot -\infty \right] \rightarrow \infty$

$\lim_{x \rightarrow -\frac{1}{e}^+} x \ln(e + \frac{1}{x}) \rightarrow \left[-\frac{1}{e} \cdot \ln(e - e^0) \right]$

$\rightarrow$ brak granicy

$x = -\frac{1}{e}$ asymptota pionowa lewostronna

b)

$$g(t) = 2t + \arctg\left(\frac{1}{2}t\right)$$

Asymptoty ukośne

$$a_L = \lim_{t \rightarrow -\infty} \frac{g(t)}{t} \rightarrow 2 + \frac{\arctg\left(\frac{1}{2}t\right)}{t} \rightarrow 2 = a_p$$

$$b_L = \lim_{t \rightarrow -\infty} (g(t) - a_L t) \rightarrow 2t + \arctg\left(\frac{1}{2}t\right) - 2t \rightarrow \arctg\left(\frac{1}{2}t\right) \xrightarrow{t \rightarrow -\infty} -\frac{\pi}{2}$$

$$b_P = \lim_{t \rightarrow +\infty} 2t + \arctg\left(\frac{1}{2}t\right) - 2t \rightarrow \arctg\left(\frac{1}{2}t\right) \xrightarrow{t \rightarrow +\infty} \frac{\pi}{2}$$

Asymptoty
ukośne

$$y = 2t - \frac{\pi}{2} \text{ dla } t \rightarrow -\infty$$

$$y = 2t + \frac{\pi}{2} \text{ dla } t \rightarrow +\infty$$

Zad. 27.2 $u, b1, f1, b1$

$$u) b(u) = u e^{u^3}$$

$$f(x) = x e^{x^3}$$

1° Dziedzina funkcji

$$D: x \in (-\infty; \infty)$$

3° Asymptoty

Własne

$$a_L = \lim_{x \rightarrow -\infty} e^{x^3} = [e^{-\infty} = \frac{1}{e^{\infty}}] \rightarrow 0$$

$$b_L = \lim_{x \rightarrow -\infty} f(x) \rightarrow x \cdot e^{x^3} \rightarrow 0$$

$$a_P = \lim_{x \rightarrow +\infty} e^{x^3} = [e^{\infty}] \rightarrow \infty$$

w. pozioma $y=0$ dla $x \rightarrow -\infty$

4° Miejsce zerowe

Z. 0Y

$$x=0 \Rightarrow y=0$$

Z. 0X

$$y=0$$

$$x \cdot e^{x^3} = 0$$

2° Granica na końcach dziedziny

$$\lim_{x \rightarrow +\infty} x \cdot e^{x^3} = [\infty \cdot \infty] \rightarrow \infty$$

$$\lim_{x \rightarrow -\infty} x \cdot e^{x^3} = [-\infty \cdot 0] \rightarrow$$

$$\rightarrow \frac{x}{e^{-x^3}} \rightarrow \left[\frac{-\infty}{-\infty} \right] \xrightarrow{H}$$

$$\xrightarrow{H} \frac{1}{e^{-x^3} \cdot (-3x^2)} \rightarrow \frac{-1}{3x^2 \cdot e^{-x^3}} \rightarrow$$

$$\rightarrow \left[\frac{-1}{\infty \cdot \infty} \right] \rightarrow \left[\frac{-1}{\infty} \right] \rightarrow 0$$

5° f'

$$f'(x) = (x \cdot e^{x^3})' =$$

$$= e^{x^3} + x \cdot e^{x^3} \cdot 3x^2 =$$

$$= e^{x^3} (1 + 3x^3)$$

$$f' > 0 \Rightarrow e^{x^3} (1 + 3x^3) > 0$$



6' f''

$$f'' = (e^{x^3}(1+3x^3))' = e^{x^3} \cdot 3x^2(1+3x^3) + e^{x^3} \cdot 9x^2 =$$

$$= e^{x^3}(9x^5 + 12x^2) = x^2 e^{x^3}(9x^3 + 12)$$

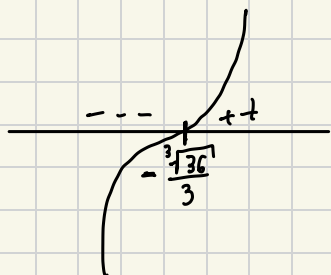
$$f'' > 0 \Rightarrow x^2 e^{x^3}(9x^3 + 12) > 0$$

$$9x^3 + 12 = 0$$

$$x^3 = -\frac{12}{9}$$

$$x^3 = -\frac{36}{27}$$

$$x = -\frac{\sqrt[3]{36}}{3}$$



7° Tabellen

x	$-\infty$		$-\frac{\sqrt[3]{36}}{3}$		0		$+\infty$
f'	+	+	+	+	+	+	+
f''	-	-	0	+	+	+	+
f	0	↗	?	↗	0	↗	$+\infty$

b1)

$$k(p) = p \cdot e^{\frac{1}{p}}$$

$$f(x) = x \cdot e^{\frac{1}{x}}$$

1° Dziedzina

$$D: x \in (-\infty; 0) \cup (0; +\infty)$$

2° Granice na końcach dziedz.

$$\lim_{x \rightarrow 0^+} x \cdot e^{\left(\frac{1}{x}\right)^{1^+}} \rightarrow [\infty \cdot 1] \rightarrow \infty$$

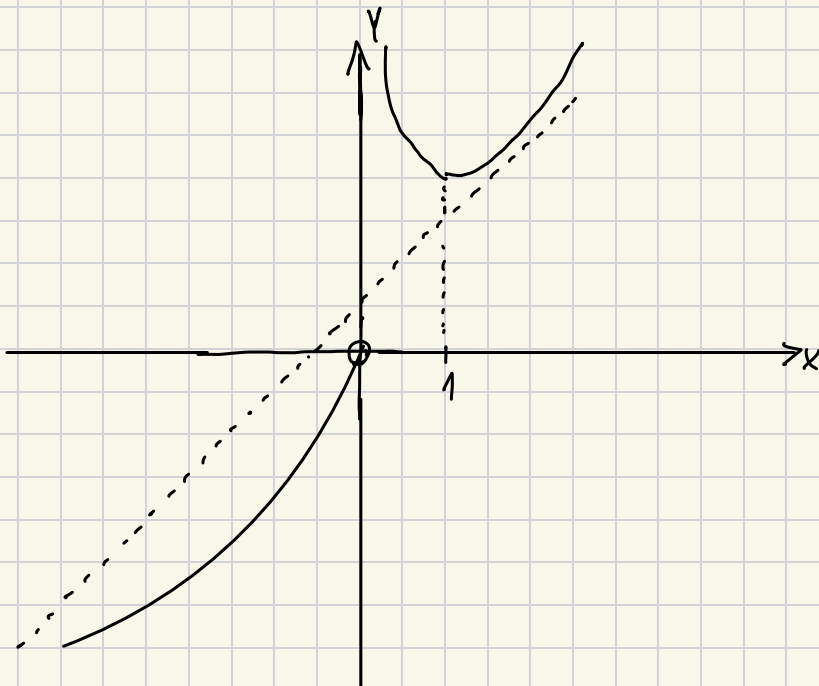
$$a_L = a_P = 1$$

$$b_L = \lim_{x \rightarrow -\infty} x(e^{\frac{1}{x}} - 1) \rightarrow \frac{e^{\frac{1}{x}} - 1}{\frac{1}{x}} \rightarrow \left[\frac{0}{0}\right]$$

$$\rightarrow \frac{e^{\frac{1}{x}} \cdot \left(-\frac{1}{x^2}\right)}{\left(-\frac{1}{x^2}\right)} \rightarrow e^{\frac{1}{x}} \rightarrow 1$$

$$b_P = \lim_{x \rightarrow +\infty} x(e^{\frac{1}{x}} - 1) \rightarrow \frac{e^{\frac{1}{x}} - 1}{\frac{1}{x}} \rightarrow \left[\frac{0}{0}\right] \rightarrow$$

$$\rightarrow \frac{e^{\frac{1}{x}} \cdot \left(-\frac{1}{x^2}\right)}{\left(-\frac{1}{x^2}\right)} \rightarrow e^{\frac{1}{x}} \rightarrow 1$$

as. ukośna. $y = x + 1$ 

$$f(x) = \ln(x^2 - 1)$$

$$f(x) = \ln(x^2 - 1)$$

$$x^2 - 1 > 0 \quad \begin{array}{c} \text{++} \\ -1 \quad 1 \end{array}$$

$$D: x \in (-\infty; -1) \cup (1; +\infty)$$

2° Granice na końcach dziedzin

$$\lim_{x \rightarrow -\infty} \ln(x^2 - 1) \rightarrow \infty$$

$$\lim_{x \rightarrow +\infty} \ln(x^2 - 1) \rightarrow \infty$$

3° Asymptoty

Pionowe

$$x = -1?$$

$$\lim_{x \rightarrow -1^-} \ln(x^2 - 1) \rightarrow -\infty$$

$$\lim_{x \rightarrow -1^+} \ln(x^2 - 1) \text{ brak}$$

$x = -1$ as. pionowa lewostronna

$$x = 1?$$

$$\lim_{x \rightarrow 1^-} \ln(x^2 - 1) \text{ brak}$$

$$\lim_{x \rightarrow 1^+} \ln(x^2 - 1) \rightarrow -\infty$$

$x = 1$ as. pionowa
prawostronna

Ukośne

$$a_L = \lim_{x \rightarrow -\infty} \frac{\ln(x^2 - 1)}{x} \rightarrow$$

$$\rightarrow \left[\frac{-\infty}{-\infty} \right] \xrightarrow{H} \frac{\frac{1}{x^2 - 1} \cdot 2x}{1} \rightarrow \frac{2x}{x^2 - 1} \rightarrow$$

$$\rightarrow \left[\frac{1}{-\infty} \right] \rightarrow 0 = a_p$$

$$b_L = \lim_{x \rightarrow -\infty} \ln(x^2 - 1) \rightarrow \infty = b_p$$

Nie ma asymptot ukośnych

$$4^o \quad z \text{ or}$$

$$x=0 \notin D \text{ brake}$$

$$z \text{ or } x$$

$$y=0$$

$$\ln(x^2-1)=0$$

$$e^0 = x^2 - 1$$

$$1 = x^2 - 1$$

$$x^2 = 2$$

$$x = \pm \sqrt{2}$$

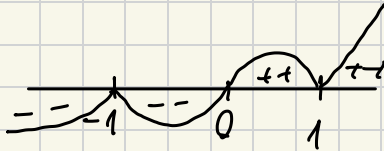
$$5^o \quad f'$$

$$f'(x) = (\ln(x^2-1))' = \frac{2x}{x^2-1}$$

$$f' > 0 \Rightarrow \frac{2x}{x^2-1} > 0 \quad || \cdot (x^2-1)^2$$

$$2x(x^2-1)^2 > 0$$

$$2x(x-1)^2(x+1)^2 > 0$$



$$6^o \quad f''$$

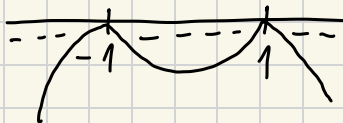
$$f''(x) = \left(\frac{2x}{x^2-1} \right)' = \frac{2(x^2-1) - 2x \cdot 2x}{(x^2-1)^2} =$$

$$= \frac{-2x^2 - 2}{(x^2-1)^2} = \frac{-2(x^2+1)}{(x^2-1)^2}$$

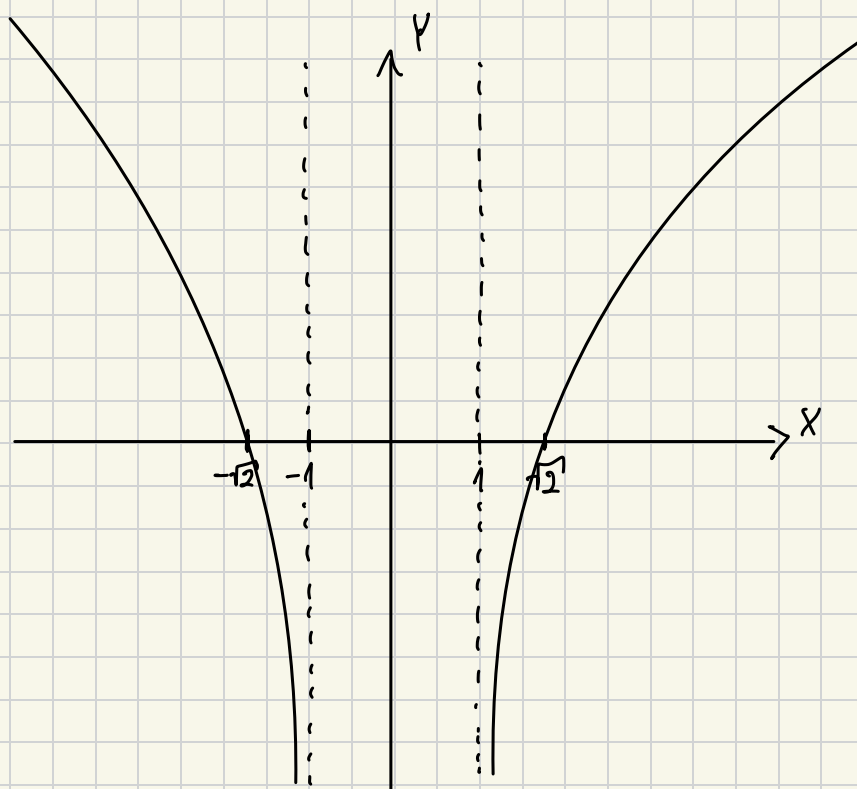
$$f'' > 0 \Rightarrow \frac{-2(x^2+1)}{(x^2-1)^2} > 0 \quad || \cdot (x^2-1)^2$$

$$-2(x^2+1)(x^2-1)^2 > 0$$

$$-2(x^2+1)(x-1)^2(x+1)^2 > 0$$



x	$-\infty$		$-\sqrt{2}$		-1		$//$		1		$\sqrt{2}$		$+\infty$
f'	$-$	$-$	$-$	$-$	$-$	$-$	$//$	$+$	$+$	$+$	$+$	$+$	$+$
f''	$-$	$-$	$-$	$-$	$-$	$-$	$//$	$-$	$-$	$-$	$-$	$-$	$-$
f	∞	$\downarrow$	0	$\downarrow$	$-\infty$	hole	$//$	hole	$-\infty$	$\uparrow$	0	$\uparrow$	∞



h1)

$$f(x) = \frac{x}{\ln x}$$

$$\ln x \neq 0 \quad x > 0$$

$$e^0 \neq x$$

$$x \neq 1$$

$$D: x \in (1; +\infty)$$

$$\lim_{x \rightarrow +\infty} \frac{x}{\ln x} \rightarrow \left[\frac{+\infty}{+\infty} \right] \xrightarrow{H} \frac{1}{\frac{1}{x}} \rightarrow x \rightarrow +\infty$$

An. pionowe

$$\lim_{x \rightarrow 1^-} \frac{x}{\ln x} \rightarrow \left[\frac{1}{0^-} \right] \rightarrow -\infty$$

$$\lim_{x \rightarrow 1^+} \frac{x}{\ln x} \rightarrow \left[\frac{1}{0^+} \right] \rightarrow +\infty$$

$x = 1$ as. pionowa asymptota

Ukośne

$$a_L = \lim_{x \rightarrow -\infty} \frac{1}{\ln x}$$

$\ln(-\infty)$ nie zdefiniowane

$$a_P = \lim_{x \rightarrow +\infty} \frac{1}{\ln x} = \left[\frac{1}{+\infty} \right] \rightarrow 0$$

$$b_P = \lim_{x \rightarrow +\infty} \left(\frac{x}{\ln x} - x \right) \rightarrow x \left(\frac{1}{\ln x} - 1 \right) \rightarrow$$
$$\rightarrow \left[+\infty \left(\frac{1}{+\infty} - 1 \right) \right] \rightarrow +\infty$$

brak as. ukośnych

Z OY

$x = 0 \notin D$ brak

$$y = 0$$

$$\frac{x}{\ln x} = 0$$

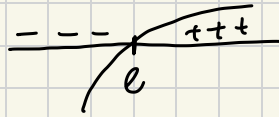
$x = 0 \notin D$ brak

$$f'(x) = \left(\frac{x}{\ln x}\right)' = \frac{1 \cdot \ln x - x \cdot \frac{1}{x}}{(\ln x)^2} = \frac{\ln x - 1}{(\ln x)^2}$$

$$f' > 0 \quad \frac{\ln x - 1}{(\ln x)^2} > 0 \quad || \cdot (\ln x)^2$$

$$\ln x - 1 > 0$$

$$\ln x > 1$$



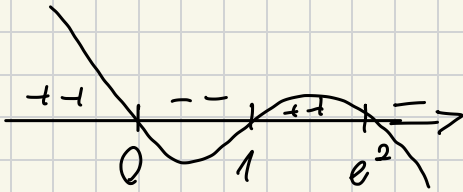
$$f''(x) = \left(\frac{\ln x - 1}{(\ln x)^2}\right)' = \frac{\frac{1}{x}(\ln x)^2 - (\ln x - 1) \cdot 2 \ln x \cdot \frac{1}{x}}{(\ln x)^2} = \frac{\frac{1}{x}[(\ln x)^2 - 2(\ln x)^2 + 2 \ln x]}{(\ln x)^2} =$$

$$f'' > 0$$

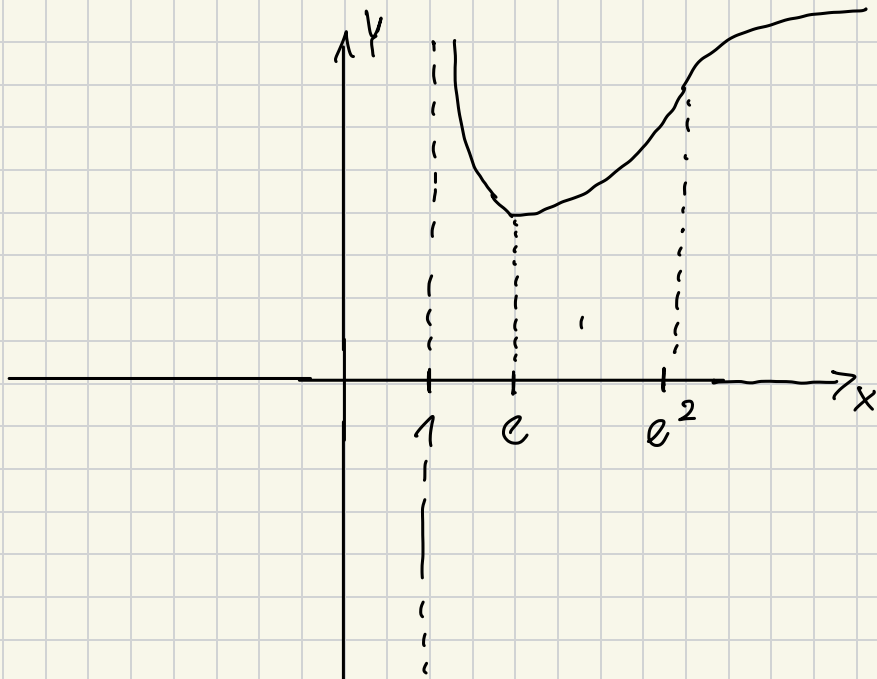
$$= \frac{\frac{\ln x}{x}(-\ln x + 2)}{(\ln x)^2} = \frac{-\ln x + 2}{x \ln x}$$

$$\frac{-\ln x + 2}{x \ln x} > 0 \quad || \cdot (x \ln x)^2$$

$$-(\ln x - 2) \cdot \ln x \cdot x > 0$$



x	1		e		e ²		+	+	+	+
f'	-	-	-	-	+	+	+	+	+	+
f''	-	+	+	+	+	+	+	-	-	-
f	+	+	?	?	?	?	?	?	?	+



$$\lim_{x \rightarrow 0^+} (\operatorname{tg} x^3)^{\frac{1}{\ln x}} \rightarrow e^{\frac{1}{\ln x} \cdot \ln \operatorname{tg} x^3} \rightarrow e^{\frac{\ln \operatorname{tg} x^3}{\ln x}} \rightarrow e^0 \rightarrow 1$$

$$\begin{aligned} \lim_{x \rightarrow 0^+} \frac{\ln \operatorname{tg} x^3}{\ln x} &\rightarrow \left[\frac{-\infty}{-\infty} \right] \xrightarrow{H} \frac{\frac{1}{\operatorname{tg} x^3} \cdot \frac{1}{\cos^3 x^3} \cdot 3x^2}{\frac{1}{x}} \rightarrow \frac{3x^3}{\sin x^3 \cdot \cos x^3} \rightarrow \\ &\rightarrow \frac{6x^3}{2 \sin x^3 \cos x^3} \rightarrow \frac{6x^3}{\sin 2x^3} \rightarrow \left[\frac{0}{0} \right] \xrightarrow{H} \frac{18x^2}{\cos 2x^3 \cdot 6x^2} \rightarrow \left[\frac{0}{1} \right] \rightarrow 0 \end{aligned}$$